

Christina Humer

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Born on April 2nd, 1996

8046 Zurich, CH

✉ christina@humer.dev

🔗 christina.humer.dev

🎓 Google Scholar

Scientific Interests

- Human-Centered AI
- AI4Materials
- Visual Analytics

Personal Interests

- Climate Change Mitigation
- klimadashboard.at
- Sports (Hiking, Climbing, Volleyball)
- Travel and Cultural Exchange
- Reading
- Gardening, Mycology, Geology

Technical Skills

- Python, Java, C#
- Tensorflow, Pytorch
- HTML, CSS, JavaScript etc.
- React, Angular, Vue, Svelte
- SQL

Languages

- German (Native)
- English (Fluent)
- Spanish (Learning)

Current Situation

Since Jun. 2025 **Postdoctoral Fellow**

ETH AI Center, Zurich

Together with the [Interactive Visualization & Intelligence Augmentation](#) (IVIA) Lab, the [Digital Chemistry Lab](#) (DCL), and the [Space Geodesy Lab](#), I work on fighting climate change with human-centered AI.

Education

2024 PhD in Engineering Sciences at Johannes Kepler University Linz, AT.

Thesis title: *Human-Centered Techniques to Foster Trust Calibration and Understanding of Machine Learning Models.*

2020 MSc in Computer Science with a focus on **Data Science** at Johannes Kepler University Linz, AT.

Thesis title: *Early Detection of Spruce Bark Beetles using Semantic Segmentation and Image Classification.*

2018 BSc in Computer Science at Johannes Kepler University Linz, AT.

Summer Term 2017 at Dublin City University, IE.

2015 Leaving Certificate at Higher Technical and Vocational College for Information Technology and Organisation in Grieskirchen, AT.

Internships and Research Exchange

Mar.-Jul. 2024 Internship at the [Mila Quebec AI Institute](#) in Montreal, CA. Use of machine learning for accelerating material discovery for climate change mitigation.

Apr.-Jul. 2023 Research Exchange at the [MIT-IBM Watson AI Lab](#) in Cambridge, MA, USA. Development of an interpretation technique for understanding multi-modal models.

Aug.-Sep. 2019 Internship at [Cubido](#) in Leonding, AT. Development of an automated workflow for preprocessing and analysing temporal data.

Scientific Publications

- Humer, C., Hinterreiter, A., Leichtmann, B., Mara, M., Streit, M. **2024** Reassuring, Misleading, Debunking: Comparing Effects of XAI Methods on Human Decisions *ACM Transactions on Interactive Intelligent Systems* 14, 3. DOI: [10.1145/366564](https://doi.org/10.1145/366564)
- Leichtmann, B., Hinterreiter, A., Humer, C., Ventura, A., Streit, M., Mara, M.. **2024** Moral reasoning in a digital age: blaming artificial intelligence for incorrect high-risk decisions *Current Psychology*. DOI: [10.1007/s12144-024-06658-2](https://doi.org/10.1007/s12144-024-06658-2)
- Humer, C., Nicholls, R., Heberle, H., Heckmann, M., Puehringer, M., Wolf, T., Luebbesmeyer, M., Heinrich, J., Hillenbrand, J., Volpin, G., Streit, M. **2024** CIME4R: Exploring iterative, AI-guided chemical reaction optimization campaigns in their parameter space *Journal of Cheminformatics* 16, 51. DOI: [10.1186/s13321-024-00840-1](https://doi.org/10.1186/s13321-024-00840-1)
- Pile, S., Lesota, O., Peter, S. D., Humer, C., Gasser, M. **2024** Spin-Wave Voices: Sonification of Nanoscale Spin Waves as an Engagement and Research Tool *International Conference on Auditory Display*. DOI: [10.21785/icad2024.024](https://doi.org/10.21785/icad2024.024)
- Humer, C., Prasad, V., Streit, M., Strobelt, H. **2023** Understanding and Comparing Multi-Modal Models *VISxAI 2023*. URL: jku-vds-lab.at/amumo
- Hinterreiter, A., Humer, C., Leichtmann, B., Mara, M., Streit, M. **2023** Of Deadly Skullcaps and Amethyst Deceivers: Reflections on a Transdisciplinary Study on XAI and Trust *VISxAI 2023*. URL: jku-vds-lab.at/hoxai-at-visxai
- Trainer, M., Humer, C., Adelberger, P., Streit, M. **2023** How Far Can Public Transport Take You? *CityVis 2023*. URL: www.cityvis.io/workshops/2023
- Hinterreiter, A., Humer, C., Kainz, B., Streit, M. **2023** ParaDime: A Framework for Parametric Dimensionality Reduction *EuroVis 2023*. DOI: [10.1111/cgf.14834](https://doi.org/10.1111/cgf.14834)
- Leichtmann, B., Hinterreiter, A., Humer, C., Streit, M., Mara, M. **2023** Explainable Artificial Intelligence improves human decision-making: Results from a mushroom picking experiment at a public art festival *Journal of Human-Computer Interaction*. DOI: [10.1080/10447318.2023.2221605](https://doi.org/10.1080/10447318.2023.2221605)
- Leichtmann, B., Humer, C., Hinterreiter, A., Streit, M., Mara, M. **2022** Effects of Explainable Artificial Intelligence on trust and human behavior in a high-risk decision task *Computers in Human Behavior* 139, 107539. DOI: [10.1016/j.chb.2022.107539](https://doi.org/10.1016/j.chb.2022.107539)
- Humer, C., Heberle, H., Montanari, F., Wolf, T., Huber, F., Henderson, R., Heinrich, J., Streit, M. **2022** ChemInformatics Model Explorer (CIME): Exploratory Analysis of Chemical Model Explanations *Journal of Cheminformatics* 14, 21. DOI: [10.1186/s13321-022-00600-z](https://doi.org/10.1186/s13321-022-00600-z)
- Humer, C., Elharty, M., Hinterreiter, A., Streit, M. **2022** Interactive Attribution-based Explanations for Image Segmentation *EuroVis 2022*. DOI: [10.2312/evp.20221130](https://doi.org/10.2312/evp.20221130)

Eckelt, K., Hinterreiter, A., Adelberger, P., Walchshofer, C., Dhanoa, V., Humer, C., Heckmann, M., Steinparz, C. A., Streit, M. **2022** Visual Exploration of Relationships and Structure in Low-Dimensional Embeddings *TVCG*. DOI: [10.1109/TVCG.2022.3156760](https://doi.org/10.1109/TVCG.2022.3156760)

Preprints

Humer, C., Rumetshofer, E., Sánchez, A., Prasad, V., Klambauer, G., Streit, M., Stobelt, H. **(2024)** Understanding and Comparing Latent Space Characteristics of Multi-Modal Models *The Journal of Visualization and Interaction*. url: <https://www.journalovi.org/2024-humer-amumo/>

Talks, Presentations, and Exhibitions

Oct. 2023 **VISxAI 2023**, Online. *Understanding and Comparing Multi-Modal Models*.

Sep. 2022 (Ongoing) **Ars Electronica Center Exhibitions** in Linz, AT. *AI Forest*.

Jun. 2022 **EuroVis 2022 Posters** in Rome, IT. *Interactive Attribution-based Explanations for Image Segmentation*.

Oct. 2021 **10th RDKit User Group Meeting**, Online. *CIME: Exploratory and Explanatory Visualization of Molecules and Chemical Models*.

Sep. 2021 **Ars Electronica Festival 2021 Exhibitions** in Linz, AT. *AI Forest*.

Awards and Scholarships

2024 **Best Paper Award** at [International Conference on Auditory Displays \(ICAD'24\)](#) .

2023 **Best Submission Award** at [Workshop on Visualization for AI Explainability \(VISxAI'23\)](#).

2023 **Marshall Plan Scholarship** for academic exchange between Austria and the U.S. granted by the Austrian Marshall Plan Foundation.

2019 **Award for excellent performance as a student** (Leistungsstipendium) granted by the Faculty of Computer Science, Johannes Kepler University Linz.

2018 **Award for excellent performance as a student** (Leistungsstipendium) granted by the Faculty of Computer Science, Johannes Kepler University Linz.